

3.2 Worker Earnings

Figure 5, Panels (a) and (b) examine earnings of chatbot adopters, distinguishing between those with and without employer encouragement (i.e., encouraged-use policies). Each group is compared to non-encouraged non-adopters—workers insulated from AI chatbots through both personal non-use and the absence of employer encouragement.

Panel (a) begins by presenting differences in means, controlling for workers’ survey occupation and pre-determined characteristics. Adopters appear to earn substantial premia of about 9% for those with employer-encouraged usage and 4% for those without. However, leveraging the panel dimension of our administrative data reveals that these earnings differentials entirely predate the arrival of AI chatbots. Indexing to November 2022, the difference-in-differences estimates in Panel (b) show no differential changes for either group of adopters following the introduction of AI chatbots. Appendix F.2 shows that these null results hold separately for workers’ hourly wages, intensive-margin hours, and extensive-margin employment. Moreover, Figure F.4 shows that the null results on worker earnings hold if we rely only on the variation in chatbot use that stems from employer policies—variation that is arguably more plausibly unrelated to individuals’ earnings trajectories.

Figure 5, Panel (b) provides three key insights. First, two years after their launch, AI chatbots have had minimal earnings impact on workers who use them. The confidence intervals of our dynamic difference-in-differences estimates rule out changes larger than 3%, and our pooled difference-in-differences estimates rule out effects larger than 2%.¹⁴ These effects are also small relative to the substantial variation in earnings changes observed for similar workers over time: the standard deviation of residual earnings changes is about 30 percentage points in December 2024. Second, these effects show no differential trends after the introduction of AI chatbots, indicating that the minimal impacts are not

¹⁴The dynamic difference-in-differences in Figure 5, Panel (b) reveal that adopters are on slightly stronger labor market trends. However, because these trends entirely predate AI chatbots, the pooled difference-in-differences in Figure 7, Panel (a) (which control for pre-trends; see Equation (3)) are precise zeros, with confidence bands ruling out effects larger than 2%.

merely a short-term phenomenon. Third, despite evidence in Section 2.2 that employer encouragement boosts work-related benefits of AI chatbots, we find no differential changes in earnings for workers whose employers encourage their chatbot use.

Heterogeneity by Intensity of Adoption. Figure 6, Panel (a) examines the earnings impacts of chatbot use by the intensity of workers’ adoption.

We first consider the effects of additional investments in employer-provided chatbots and training, offering insights into the impacts of AI chatbot use when firms pursue a more comprehensive strategy. Yet even when the “full package” of encouraged use, enterprise chatbots, and training substantially increases reported benefits (Section 2.2), we find no detectable impact on adopter earnings.

Next, we consider whether the null effects reflect the limited intensity of individual use. In particular, even if most adopters do not use the tools daily, a meaningful fraction of workers report more substantial benefits—such as daily time savings exceeding one hour. Turning to workers who use AI chatbots daily or who adopted within the first year after ChatGPT’s launch, we find no differential earnings changes since the arrival of AI chatbots, despite their substantial engagement with the tools. Similarly, among workers who report greater benefits from chatbot use—time savings exceeding one hour per day, higher-quality output, and enhanced creativity—we find no differential earnings changes in the administrative data. Taken together, although “super users” (those with the most intensive adoption or highest reported benefits) certainly appear in our data, even their more intensive engagement with AI chatbots has not translated into detectable earnings gains.

Finally, we consider workers who report new workloads from AI chatbots—and new AI-related tasks in particular. Even if chatbot use does not directly affect adopter earnings, it could influence them indirectly through the creation of new work. Yet neither workers with new workloads nor those with new AI-related tasks have experienced differential earnings changes since the arrival of AI chatbots.

Occupational Heterogeneity. The overall zero effects could mask important heterogeneity across occupations. For instance, since teachers are covered by collective bargaining with centralized wage setting, it may be less surprising that individual productivity gains do not translate into higher earnings. By contrast, occupations such as marketing and software development feature decentralized wage setting, allowing more flexibility to adjust pay based on individual productivity.¹⁵ Appendix C.1 provides an overview of the wage-setting systems in Denmark, showing rich variation across our study occupations. Occupational heterogeneity may also arise from whether chatbots are used primarily for core versus peripheral tasks within an occupation (Autor and Thompson, 2025).

Figure 7, Panel (a) shows the estimated effects of AI chatbots on worker earnings across the 11 occupations in our sample. Confidence bands are wider for these occupation-specific estimates and should therefore be interpreted with caution. Nonetheless, we find no statistically significant effects in any occupation, and point estimates are generally near zero. Even for our most contrasting groups—software developers and marketers, with more flexible wages and larger reported time savings, versus teachers, with centralized wage-setting and smaller reported savings—we find no systematic differences in earnings impacts.

Discussion. Our administrative data show that users of AI chatbots have not experienced differential changes in earnings or hours following the arrival of these tools. Even among workers who use chatbots daily and report substantial benefits, we find no net changes—not even a break in trends. These patterns echo those documented for computer use in the 1980s by Krueger (1993) and critiqued by DiNardo and Pischke (1997): although chatbot use is correlated with sizable cross-sectional earnings premia, these differentials vanish once accounting for positive selection.

Are the null wage effects of chatbot use theoretically surprising? Competitive labor markets (e.g., our framework in Appendix A.2) imply that workers should be compensated

¹⁵Furthermore, marketing professionals and software developers are exactly the occupations with the highest reported use and productivity gains from the tools (Appendix E.3).

for productivity gains that are not driven by employer initiatives (Acemoglu and Pischke, 1998; Becker, 1964). From this perspective, it is surprising that individual-specific benefits from AI chatbots—such as those among non-encouraged or high-use, high-benefit adopters—do not translate into higher earnings. On the other hand, if employers do not endorse or invest in the tools, they may be reluctant to reward their use. From this perspective, it is surprising that adopters supported by employer initiatives have also not experienced earnings gains. One possibility is that workers value chatbot access as a nonpecuniary amenity, accepting unchanged wages in exchange for easier or more engaging work (Mas, 2025). The null effects also persist across occupations with very different wage-setting institutions—from centralized collective bargaining for teachers to decentralized individual negotiation for software developers and marketers—suggesting that wage rigidity alone cannot explain the pattern. Indeed, the standard deviation of residual earnings changes in Equation (2) is about 30 percentage points in December 2024.

Another possibility is that workers overestimate the benefits they derive from these tools (Becker et al., 2025; Edelman, Ngwe and Peng, 2023).¹⁶ A key contribution of our study is to link these self-reports to third-party administrative data, allowing us to move beyond purely self-reported effects. That said, inflated self-reports are unlikely to be the full explanation: Section 3.5 shows that even workers themselves recognize that AI chatbots have not materially affected their earnings.

3.3 Occupational Mobility

Figure 5, Panel (c) shows that adopters have experienced greater occupational mobility since the arrival of AI chatbots. We measure occupational mobility by tracking hours worked in each worker’s most recent occupation as of December 2024. If adopters are more likely to switch into new occupations, their hours in that new occupation will rise relative to non-adopters, who are more likely to remain in their original occupation. Because the figure reports difference-in-differences estimates indexed to November 2022, the estimates

¹⁶Consistent with experimental evidence from Becker et al. (2025), Appendix E.4 shows that workers who report greater time savings from chatbots are also more likely to report taking on new tasks.